

EXACT 2026 — Solution Description

Team: **superNB** | IEEE IJCNN 2026 — The 2nd International XAI Challenge for Transparent Educational Question-Answering

1. Datasets Used

Only the two official EXACT 2026 datasets were used. No external, crawled, or synthetic data was used for training or tuning.

- **Physics_Problems_Text_Only.csv** (official EXACT release) — 1,352 Type 2 physics problems; all 1,352 used for pipeline development and evaluation (80/20 train/test split for parser-template development). Sample: "Two point charges, $q_1 = 0.5 \text{ nC}$ and $q_2 = -0.5 \text{ nC}$, are placed at points A and B, 6 cm apart in air. What is the magnitude of the electric field strength at the midpoint?" → answer 10000, unit V/m.
- **Logic_Based_Educational_Queries.json** (official EXACT release) — Type 1 logic records expanded to 804 questions (642 train / 162 validation) for the retained reasoner. Sample: premises "A student who has completed at least 120 credits is eligible for graduation"; "Student A has completed 118 credits", question "Is Student A eligible for graduation?" → "No".

2. Approach and Method

A single public **POST /predict** endpoint receives the unified EXACT schema and routes each query by its **type** field to one of two pipelines. Every response contains the answer, ASCII unit, a natural-language explanation, premises_used indices, and structured reasoning steps.

- **Type 1 (logic)**: question grouping (mcq/judgment × logic shape) → BGE-RAG semantic memory (BAAI/bge-small-en-v1.5 retriever) → modal-abductive possible-world traces (support / contradiction / uncertainty worlds) → SSM world-model latent states injected into a compact Transformer reasoner that ranks answer candidates. Choice questions are answered by this retained model (validation accuracy 0.827); free-form number/text questions are answered by the vLLM-served LLM with premise-grounded FOL reasoning steps.
- **Type 2 (physics)**: deterministic-first staged pipeline — Stage 0 rule-based parsing (quantities, target, domain; LLM recovery only for failed parses) → formula retrieval over a curated physics formula library (beam search) → deterministic symbolic solving (SymPy) with per-step verification → trace diagnosis and guarded repair → final-answer validation → response assembly. The LLM is called only when the deterministic path fails; correct deterministic traces never touch the LLM. Full-dataset objective answer accuracy: **97.58%** (1251/1282); average inference latency <2 s per query, worst case <30 s (within the 60 s limit).

3. Model Size Calculation

Component	Type	Parameters	Serving
Qwen/Qwen3-8B-AWQ (id: qwen3-8b-awq)	LLM (the only one)	8B-class (dense)	vLLM, /v1/models verifiable
BAAI/bge-small-en-v1.5	Embedding retriever (non-generative)	33M	in-process
Retained Type-1 reasoner (SSM+Transformer)	Custom non-LLM model	<10M	in-process
SymPy / rule parser / formula library	Non-LLM tools	—	in-process

The pipeline contains exactly **one** LLM: Qwen/Qwen3-8B-AWQ (8B-class, AWQ 4-bit weights), served by a single vLLM instance and shared by all LLM-dependent components (Type 1 free-form reasoning, Stage 0 parse recovery, Type 2 solve/repair fallback). At any single moment during inference the total parameters of all LLMs loaded and running is 8B-class ≤ 8B limit (per Q3 of the Official Q&A). No second LLM is ever resident: the embedding retriever and the retained reasoner are non-generative, non-LLM models, and symbolic tools do not count toward the limit. The committee can verify the served model at the /v1/models URL listed in urls.txt.

Endpoints: prediction <https://predict.exact2026tsupernb.xyz/predict> | model info <https://vllm.exact2026tsupernb.xyz/v1/models>